Ling 285
Human Language and Technology
Lecture 11

Masses and Springs
Theory and Prediction

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Pop Quiz

Theory, Prediction, and Experiment

- The Scientific Method depends on the statement of very clear theories about how the world is like, usually in the form of differential equations.
- We then solve these differential equations, and the result is a *prediction* of what the world is like when we actually measure it!
- Theory: The vocal folds are a membrane, so when it vibrates, the wave of vibration will have equidistant peaks in the spectrum (which we call harmonics).
- The differential equations are many of the type we have already seen, but slightly more complicated
- Prediction: if I measure any of your laryngeal vibration signals, they will have a spectrum with equidistant harmonics.
- Let’s do an experiment to find out if this is true!!!!!